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REVIEW

Clinical outcomes of surgical laser interventions and adjunctive photobiomodulation in osseous crown lengthening: A systematic review

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Abstract

To synthesize the available clinical evidence on laser-based interventions in osseous crown-lengthening procedures, considering surgical laser techniques and adjunctive photobiomodulation separately. A systematic search of PubMed/MEDLINE, Scopus, ScienceDirect, and Web of Science was conducted from database inception to June 2025. We included clinical studies involving adults undergoing osseous crown lengthening with laser-assisted surgery or postoperative photobiomodulation. Three reviewers independently screened titles and abstracts, assessed potentially eligible full-text articles, and extracted data. Risk of bias was assessed at the outcome level using RoB 2 for randomized studies and ROBINS-I for the nonrandomized comparative study. Because of substantial clinical and methodological heterogeneity, we evaluated findings using a structured narrative synthesis rather than meta-analysis. We included five clinical studies published between 2021 and 2024. Three evaluated lasers used during the surgical procedure, while two evaluated adjunctive photobiomodulation following conventional surgery. Some studies reported reduced intraoperative bleeding and improved surgical visibility with surgical laser techniques, although they did not consistently assess these outcomes using standardized quantitative measures. Periodontal and gingival-margin outcomes were generally comparable with conventional procedures. Findings regarding postoperative pain were inconsistent, with benefits reported in some but not all studies. Across two small studies using different irradiation protocols, adjunctive photobiomodulation was associated with reduced early postoperative pain at selected assessment times; only one study reported lower analgesic consumption. Among the four randomized studies, one assessed result was judged to have a high overall risk of bias and three raised some concerns; none was judged to have a low overall risk of bias. The nonrandomized comparative study was judged to have a serious risk of bias. Some studies descriptively reported potential intraoperative advantages with surgical laser techniques, while adjunctive photobiomodulation may reduce early postoperative pain at selected assessment times. Given the small and heterogeneous evidence base and its methodological limitations, current evidence remains insufficient to establish superiority over conventional osseous crown-lengthening surgery or to identify an optimal laser protocol.

Clinical trial number: not applicable.

Keywords Crown lengthening · Dental lasers · Photobiomodulation · Periodontal surgery · Postoperative pain · Gingival margin

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Introduction

Crown lengthening (CL) is a periodontal surgical procedure used to increase clinical crown exposure for esthetic indications, such as excessive gingival display (EGD), and for restorative indications, including subgingival caries or fractures and inadequate ferrule height [1, 2]. The present review was restricted to osseous CL, defined as procedures requiring alveolar bone removal or recontouring [3]; soft-tissue-only gingivectomy or crown lengthening without osseous resection was outside the scope of the review. Within this clinical setting, two distinct applications of laser technology were evaluated: lasers used as part of the surgical procedure and postoperative photobiomodulation (PBM) applied after conventional osseous CL surgery [4]. Because these applications address different clinical questions, they were considered within the same osseous CL clinical context but were prespecified and synthesized as separate evidence streams rather than treated as a single intervention.

The conventional method for CL involves scalpel-based flap elevation combined with rotary or manual osseous resection. Owing to its open-flap design, this approach provides direct visualization of the surgical field and precise control over bone removal, while remaining cost-effective [5, 6]. Conventional osseous CL may be accompanied by changes in gingival-margin position during healing [1, 7–9]. It has also been associated with greater postoperative bleeding and pain relative to minimally invasive approaches [10]. However, these outcomes may depend on the extent of surgery, flap design, amount of osseous resection, and patient-related factors. These potential complications, along with the drawbacks of conventional techniques more broadly, have contributed to interest in alternative or adjunctive surgical approaches, including laser-based techniques [11].

Advances in laser technology have expanded the range of surgical modalities available in periodontology [6, 12]. Different laser systems are selected based on their tissue interaction characteristics. Erbium lasers, including erbium-doped yttrium aluminum garnet (Er: YAG) and erbium, chromium-doped yttrium scandium gallium garnet (Er, Cr: YSGG), can ablate both soft and hard tissues and are therefore applicable in osseous CL procedures. In contrast, carbon dioxide (CO_2), neodymium-doped yttrium aluminum garnet (Nd: YAG), and diode lasers are primarily used for soft-tissue management [12, 13]. In addition, low-level laser therapy (LLLT), also referred to as photobiomodulation (PBM), has been introduced as an adjunctive modality to support postoperative healing [4]. These characteristics position lasers as potentially valuable tools in periodontal therapy.

Recent systematic reviews have addressed narrower aspects of laser use in periodontal crown-lengthening procedures. Hamadah et al. [14] evaluated laser-assisted gingivectomy in comparison with conventional surgical methods, whereas Elafifi Ebeid et al. [15] focused specifically on flapless Er: YAG- and Er, Cr: YSGG-assisted crown lengthening. These reviews did not provide an integrated assessment of the broader clinical evidence on osseous CL encompassing different surgical laser systems together with postoperative PBM, nor did they explicitly separate these two clinically distinct applications within the same evidence framework. Consequently, uncertainty remains regarding the comparative clinical outcomes of surgical laser techniques, the adjunctive effects of PBM after conventional osseous CL, and the consistency of findings across pain, periodontal, gingival-margin, and patient-reported outcomes.

Although surgical laser interventions and adjunctive PBM differ in mechanism and timing, both are used within the same clinical pathway of osseous crown-lengthening surgery and may influence clinically relevant perioperative outcomes. Surgical lasers modify the operative technique itself through soft-tissue incision and/or osseous recontouring, whereas PBM is administered after conventional surgery as a postoperative adjunct. Because these interventions address distinct clinical questions, they were not combined analytically; instead, they were evaluated as two separate evidence streams within the broader clinical context of osseous CL. Accordingly, this systematic review aimed to synthesize the available clinical evidence on laser-based interventions in osseous CL, considering surgical laser interventions and adjunctive PBM separately. Outcomes of interest included intraoperative performance, postoperative pain and analgesic use, healing and periodontal outcomes, adverse events, patient satisfaction, and gingival-margin stability.

Materials and methods

Study protocol

This systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 statement (PRISMA 2020) [16]. The primary research question was: “Among adults undergoing osseous CL surgery, what clinical and patient-reported outcomes are associated with laser-based surgical interventions or adjunctive PBM compared with conventional surgery, sham treatment, or no adjunctive laser treatment?” The study protocol was registered in the PROSPERO International Prospective Register of Systematic Reviews (registration number: CRD420251086585).

Eligibility criteria

Eligibility criteria were defined according to the population, intervention, comparison, outcomes, and study-design components of the PICOS framework, with the complete inclusion and exclusion criteria presented in Table 1.

Studies were eligible if they investigated adults undergoing CL surgery involving confirmed osseous resection or bone recontouring. Eligible laser applications included: (1) lasers used as surgical instruments for osseous recontouring and/or soft-tissue incision and (2) PBM applied adjunctively after conventional osseous CL surgery. We included randomized trials, nonrandomized comparative clinical studies, and prospective uncontrolled clinical studies. We included uncontrolled studies only for descriptive assessment of feasibility, healing, gingival-margin stability, and adverse events, and did not use them to infer comparative effectiveness. Only full-text English-language studies were included to ensure consistent and reliable full-text assessment and data extraction within the available review resources.

We excluded studies that evaluated gingivectomy or soft-tissue CL without osseous resection, orthodontic extrusion, periodontal pocket-reduction surgery, nonhuman or laboratory models, isolated case reports, retrospective case series, reviews, protocols, conference abstracts, commentaries, editorials, or letters.

Information sources and search strategy

We conducted a comprehensive literature search in PubMed/MEDLINE, Scopus, Web of Science, and ScienceDirect from database inception to June 23, 2025. The same predefined Boolean search concept was adapted to

Table 1 Consolidated PICOS-based inclusion and exclusion criteria for study eligibility

PICOS/domain	Inclusion criteria	Exclusion criteria
Population (P)	Human adults aged ≥ 18 years undergoing osseous crown-lengthening surgery for esthetic or functional/restorative indications	Soft-tissue-only crown lengthening or gingivectomy without osseous resection; orthodontic extrusion or orthodontic crown lengthening; animal, laboratory, or in vitro studies
Intervention (I)	Any laser modality used during or adjunctive to osseous crown lengthening, including: (1) laser-assisted soft-tissue incision combined with osseous recontouring; (2) laser-assisted osseous recontouring or osteotomy; or (3) postoperative photobiomodulation following conventional osseous crown-lengthening surgery	Lasers used solely for soft-tissue procedures without osseous crown lengthening; laser debridement or disinfection unrelated to crown-lengthening surgery; laser use in unrelated periodontal or restorative procedures
Comparator (C)	Conventional scalpel and/or rotary osseous crown-lengthening surgery; sham laser; no adjunctive laser treatment or standard care. No comparator was required for eligible prospective uncontrolled clinical studies	Comparators unrelated to osseous crown-lengthening surgery or to the intervention under evaluation
Outcomes (O)	At least one quantitative clinical or patient-reported outcome, including intraoperative bleeding or surgical visibility, clinical crown length, operative time, postoperative pain, analgesic consumption, healing or periodontal parameters, gingival-margin stability or tissue rebound, patient satisfaction, or adverse events	Studies reporting only histological, molecular, or radiographic endpoints without clinical or patient-reported outcomes
Study design (S)	Randomized clinical trials, including parallel-group and split-mouth designs; nonrandomized comparative clinical studies; prospective uncontrolled clinical studies	Isolated case reports; retrospective case series; cross-sectional studies without an eligible intervention; retrospective studies without prospective clinical follow-up; reviews and protocols
Follow-up	Minimum follow-up of ≥ 1 week	Follow-up shorter than 1 week or not reported
Language	Full-text publication in English	Non-English publications without an English full text
Publication status	Peer-reviewed full-text articles	Conference abstracts; abstracts without an available full-text article; unpublished data; theses; technical reports; editorials; commentaries; letters

the corresponding search fields available in each source, as detailed in Table 2. Reference lists of eligible studies and relevant reviews were also manually screened to identify additional eligible publications.

The search strategy included the following terms: ("crown lengthening" OR "clinical crown lengthening" OR "esthetic crown lengthening" OR "gingivectomy") AND (laser OR "laser therapy" OR "dental lasers" OR "Er: YAG" OR "Er, Cr: YSGG" OR "ErCr: YSGG" OR "diode laser" OR "photobiomodulation" OR "low-level laser therapy" OR "LLLT"). The term "gingivectomy" was included to maximize search sensitivity because some osseous CL studies may be indexed or described under this term; studies involving soft-tissue gingivectomy without osseous resection were subsequently excluded during screening. Table 2 presents the detailed search strategies and numbers of records retrieved from each database.

Reference lists of eligible full-text studies and relevant reviews were manually screened to identify additional published studies.

Table 2 Electronic search strategies and records retrieved

Source	Search strategy as executed	Search field	Final search date	Records retrieved
PubMed/MEDLINE	("crown lengthening"[Title/Abstract] OR "clinical crown lengthening"[Title/Abstract] OR "esthetic crown lengthening"[Title/Abstract] OR gingivectomy[Title/Abstract]) AND (laser[Title/Abstract] OR "laser therapy"[Title/Abstract] OR "dental lasers"[Title/Abstract] OR "Er: YAG"[Title/Abstract] OR "Er, Cr: YSGG"[Title/Abstract] OR "ErCr: YSGG"[Title/Abstract] OR "diode laser"[Title/Abstract] OR photobiomodulation[Title/Abstract] OR "low-level laser therapy"[Title/Abstract] OR LLLT[Title/Abstract])	Title/Abstract	June 23, 2025	134
Scopus	TITLE-ABS("crown lengthening" OR "clinical crown lengthening" OR "esthetic crown lengthening" OR gingivectomy) AND TITLE-ABS(laser OR "laser therapy" OR "dental lasers" OR "Er: YAG" OR "Er, Cr: YSGG" OR "ErCr: YSGG" OR "diode laser" OR photobiomodulation OR "low-level laser therapy" OR LLLT)	Title and abstract	June 23, 2025	315
Web of Science Core Collection	(TI=("crown lengthening" OR "clinical crown lengthening" OR "esthetic crown lengthening" OR gingivectomy) OR AB=("crown lengthening" OR "clinical crown lengthening" OR "esthetic crown lengthening" OR gingivectomy)) AND (TI=(laser OR "laser therapy" OR "dental lasers" OR "Er: YAG" OR "Er, Cr: YSGG" OR "ErCr: YSGG" OR "diode laser" OR photobiomodulation OR "low-level laser therapy" OR LLLT) OR AB=(laser OR "laser therapy" OR "dental lasers" OR "Er: YAG" OR "Er, Cr: YSGG" OR "ErCr: YSGG" OR "diode laser" OR photobiomodulation OR "low-level laser therapy" OR LLLT))	Title and abstract	June 23, 2025	174
ScienceDirect	("crown lengthening" OR "clinical crown lengthening" OR "esthetic crown lengthening" OR gingivectomy) AND (laser OR "laser therapy" OR "dental lasers" OR "Er: YAG" OR "Er, Cr: YSGG" OR "ErCr: YSGG" OR "diode laser" OR photobiomodulation OR "low-level laser therapy" OR LLLT)	Title, abstract, or author-specified keywords	June 23, 2025	22

Total records identified: 645

No additional search filters were applied

Study selection, data collection, and measurements

All retrieved records were imported into EndNote 2020, and duplicates were removed before screening. Three reviewers independently screened the titles and abstracts of the retrieved records. Potentially eligible articles were subsequently assessed independently in full text by the same three reviewers. The reviewers resolved disagreements at either stage through discussion and, when necessary, consultation with a fourth author.

The same three reviewers independently extracted data using a standardized data-extraction form developed for this review. Extracted items included study design, participant and sample characteristics, intervention and comparator details, laser type and wavelength, irradiation parameters, procedural characteristics, outcome measures, assessment time points, numerical results, and follow-up duration. Before full data extraction, the extraction form and selected variables were checked on a subset of the included studies and reviewed with the senior author to ensure consistency and completeness. The independently extracted data were subsequently cross-checked, and disagreements were resolved through discussion with the fourth author. The final extracted data and evidence table were also reviewed for accuracy and consistency.

Data synthesis

Clinical and methodological heterogeneity was assessed by comparing study design, participant characteristics and clinical indications, intervention purpose, laser system and wavelength, irradiation parameters, surgical technique, comparator, outcome definition, assessment method, and follow-up duration. The included studies evaluated conceptually distinct applications of laser technology, including lasers used during the surgical procedure and photobiomodulation used as an adjunct after conventional osseous crown-lengthening surgery. Quantitative pooling across these interventions was therefore considered clinically inappropriate, and a structured narrative synthesis was performed.

Studies were grouped into two primary evidence streams: (1) surgical laser interventions, in which laser technology was used for soft-tissue incision and/or osseous recontouring, and (2) adjunctive photobiomodulation, in which laser irradiation was applied following conventional osseous crown-lengthening surgery. Findings from randomized and nonrandomized comparative studies were synthesized narratively within the relevant intervention category and interpreted in light of study design and risk of bias.

Within each evidence stream, findings were organized according to predefined clinical outcome domains, including intraoperative bleeding and surgical visibility, operative time, postoperative pain and analgesic consumption, wound healing and periodontal parameters, adverse events, patient satisfaction, and gingival-margin stability.

For each outcome domain, findings were interpreted in relation to study design, sample size, intervention and comparator characteristics, direction and magnitude of the reported effect, measures of variability or precision when available, assessment time points, follow-up duration, and risk of bias. Clinical heterogeneity in laser type, wavelength, treatment protocol, surgical technique, comparator, and outcome definition was considered when evaluating the consistency of findings across studies.

Where formal effect estimates were available, these were considered together with the corresponding measures of variability and statistical uncertainty. Where formal effect estimates were not reported, findings were described narratively according to the reported direction and magnitude of between-group differences, together with sample size, assessment time points, measures of variability, and statistical uncertainty where available. Interpretation was not based solely on statistical significance.

The synthesis was reported in accordance with the PRISMA 2020 statement and the Synthesis Without Meta-analysis (SWiM) reporting guideline [17]. Although quantitative synthesis was initially planned, substantial clinical and methodological heterogeneity precluded meaningful pooling, and the change to structured narrative synthesis was documented as a protocol amendment.

Risk-of-bias assessment

Three authors independently assessed the risk of bias of the included studies using design-appropriate tools. For the four randomized comparative studies, the Revised Cochrane Risk of Bias Tool for Randomized Trials (RoB 2; version 22 August 2019) was used. RoB 2 assessments were performed for a specific result contributing to the review synthesis, defined by the relevant outcome and assessment time point, and focused on the effect of assignment to intervention. Table 4 specifies the assessed result for each randomized study.

RoB 2 evaluates five domains: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in measurement of the outcome, and bias in selection of the reported result. Overall judgments were categorized as low risk, some concerns, or high risk according to the RoB 2 algorithm [18]. The single nonrandomized comparative study was assessed using the Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I) tool [19]. We assessed the specific gingival-margin outcome contributing to the review synthesis. ROBINS-I evaluates bias due to

confounding, participant selection, classification of interventions, deviations from intended interventions, missing data, outcome measurement, and selection of the reported result. Overall risk of bias was classified as low, moderate, serious, or critical, with "no information" used where appropriate.

We assessed risk of bias using the signaling questions and decision rules specified in the respective tools. We resolved disagreements through discussion and, when necessary, consultation with a fourth reviewer. Detailed signaling-question responses and supporting judgments for the RoB 2 and ROBINS-I assessments are provided in Supplementary Tables S1 and S2, respectively.

Results

Study selection

The database search identified 645 records. After removal of 233 duplicates, 412 records underwent title and abstract screening. We assessed 28 full-text articles for eligibility; 23 were excluded (Fig. 1). Five clinical studies met the eligibility criteria and were included in the systematic review. We classified studies by whether lasers were used during the surgical procedure or as adjunctive PBM following conventional surgery.

Study characteristics

Five clinical studies published between 2021 and 2024 included adult patients undergoing osseous CL procedures for esthetic or restorative indications. The laser systems used were Er, Cr: YSGG ($n=2$), single-wavelength diode lasers ($n=2$), and a dual-wavelength diode approach ($n=1$). Three studies evaluated lasers used during the surgical procedure for osseous recontouring and/or soft-tissue incision, whereas two studies evaluated postoperative PBM as an adjunct to conventional osseous CL surgery. Table 3 presents the study protocols, participant numbers, laser parameters, and key findings of the included studies. Detailed participant and clinical characteristics are provided in Supplementary Table

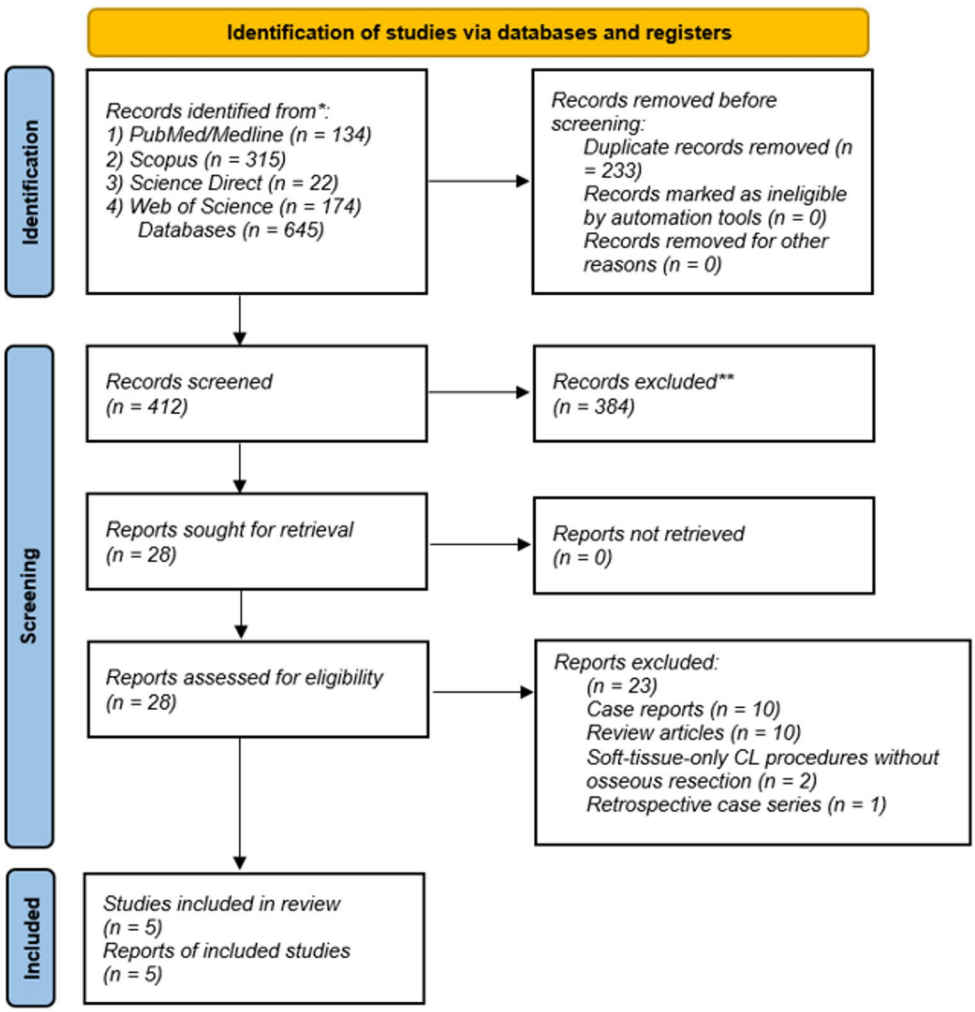


Fig. 1 PRISMA 2020 flow diagram showing study identification and selection, including records retrieved, screened, assessed in full text, excluded, and included

Table 3 Descriptive characteristics, laser parameters, and key findings of the included studies

Study ID	Study Protocol	Participants (*n*)	Laser Type	Wavelength (nm)	Laser Protocol	Energy (mJ)/ Energy Density (J/cm²)	Power (W)	Mode of Irradiation	Key findings
Sadighi et al. [22]	L vs. P (Par RCT)	20	Diode GaAlAs	860	Healing	NR/3	0.1	CW	No significant difference in pain levels on Days 1 or 7 ($p>0.05$). Lower pain intensity was observed in the L group on Day 3 ($p=0.003$). No significant difference in analgesic consumption was observed between groups.
Abdelhafez et al. [20]	L vs. CS (SM RCT)	18	Diode	940	Gingivectomy	NR/NR	2	CW	No significant difference in keratinized gingival width at 6 months ($p=0.862$). No significant difference in crown length was observed ($p>0.05$), except for the 1st upper right tooth at the 6-month recall (scalpel > laser; $p=0.039$). No significant difference in postoperative pain was observed ($p>0.05$).
Altayeb et al. [21]	L-assisted FL vs. OF (Par RCT)	36	Er, Cr: YSGG	2780	Gingivectomy	NR/57	3	Pulsed (50 Hz)	No significant difference in gingival index was observed between the OF and FL groups. Mean gingival tissue rebound at 9 months was 0.26 ± 0.29 mm in the FL group and 0.25 ± 0.32 mm in the OF group, with no significant between-group difference ($p=0.742$); participants with a thick periodontal phenotype reported greater rebound. Supracrestal gingival-tissue dimensions differed significantly at 9 months compared with immediately after surgery in both groups.
					Flap incision	NR/106	2.5	Pulsed (50 Hz)	
					Ostectomy	NR/148	4	Pulsed (20 Hz)	
					Outer-crest smoothing	NR/95	3	Pulsed (30 Hz)	
Tianmitrapap et al. [11]	L-assisted FL vs. CS (nonrandomized comparative study)	25	Er, Cr: YSGG	2780	Gingivectomy	NR/NR	1.5	CW	In treated teeth, no significant between-group difference was observed in relative gingival-margin position, whereas relative bone level was positioned more apically in the CS group. In adjacent teeth, relative gingival-margin position and relative bone level were more apically positioned in the CS group. Pain intensity was lower in the L group.
					Ostectomy	NR/NR	3.5	CW	
Golpasandhagh et al. [23]	L vs. P (SM RCT)	20	Dual L	980 and 810	Healing	NR	0.5	CW	Pain intensity and analgesic consumption were significantly lower in the L group than in the P group.

Note: Where multiple laser protocols or procedural stages are listed within a study row, they represent different applications within the same study sample and should not be interpreted as separate studies or independent cohorts. Participant counts refer to individual study participants. For split-mouth studies, the numbers of treated teeth/sites and within-participant allocations are reported separately in Supplementary Table S3

Abbreviations: L, laser; P, placebo; CS, conventional surgery; FL, flapless; OF, open-flap; Er, Cr: YSGG, erbium, chromium-doped yttrium scandium gallium garnet; CW, continuous wave; Par RCT, parallel randomized clinical trial; GaAlAs, gallium aluminum arsenide; SM RCT, split-mouth randomized clinical trial; NR, not reported

S3, while detailed study-level numerical outcome data are provided in Supplementary Table S4.

Risk of bias

Outcome-specific RoB 2 assessments were conducted for the four randomized comparative studies (Table 4; Fig. 2). One assessed result was judged to have a high overall risk of bias, while the remaining three raised some concerns; none was judged to have a low overall risk of bias. The principal concerns involved the randomization process, missing outcome data, outcome measurement, and uncertainty regarding selection of the reported result. Detailed signaling-question responses and supporting judgments are provided in Supplementary Table S1.

We assessed the single nonrandomized comparative study using ROBINS-I (Table 5; Fig. 3). We judged the overall risk of bias as serious, primarily because treatment

Table 4 Outcome-specific risk-of-bias judgments for the randomized comparative studies using the Revised Cochrane Risk of Bias Tool for Randomized Trials (RoB 2)

Study	Specific result assessed	D1 Randomization	D2 Deviations	D3 Missing data	D4 Outcome measurement	D5 Selection of reported result	Overall
Sadighi et al. [22]	Postoperative pain intensity (VAS), Day 3, PBM vs. placebo	Some concerns	Low	Low	Low	Some concerns	Some concerns
Abdelhafez et al. [20]	Clinical crown length at 6 months, diode laser vs. scalpel	Some concerns	Low	Low	Some concerns	Some concerns	Some concerns
Altayeb et al. [21]	Gingival tissue rebound at 9 months, open-flap vs. flapless laser-assisted ECL	High	Some concerns	Some concerns	Some concerns	Some concerns	High
Golpasandhagh et al. [23]	Postoperative pain intensity (VAS), Day 1, PBM vs. placebo	Low	Low	Low	Low	Some concerns	Some concerns

D1, bias arising from the randomization process; **D2**, bias due to deviations from intended interventions; **D3**, bias due to missing outcome data; **D4**, bias in measurement of the outcome; **D5**, bias in selection of the reported result. We categorized risk-of-bias judgments as low risk, some concerns, or high risk. Assessments focused on the effect of assignment to intervention and were performed for the specific result identified for each randomized study. Detailed signaling-question responses and supporting judgments are provided in Supplementary Table S1. (Assessments used RoB 2, version 22 August 2019.)



Fig. 2 Outcome-specific RoB 2 assessment of the randomized comparative studies: (**a**) traffic-light plot; (**b**) summary plot

Table 5 Outcome-specific risk-of-bias assessment of the nonrandomized comparative study using ROBINS-I

Study	Result assessed	D1	D2	D3	D4	D5	D6	D7	Overall
Tianmitrapap et al. [11]	Change in relative gingival-margin position from immediately post-surgery to 3 months in treated teeth	Serious	Low	Low	Low	Low	Moderate	Low	**Serious**

D1, bias due to confounding; **D2**, bias in selection of participants into the study; **D3**, bias in classification of interventions; **D4**, bias due to deviations from intended interventions; **D5**, bias due to missing data; **D6**, bias in measurement of outcomes; **D7**, bias in selection of the reported result. ROBINS-I judgments were categorized as low, moderate, serious, or critical risk of bias, with an additional category of no information where applicable. (The assessment used the original 2016 ROBINS-I tool.)

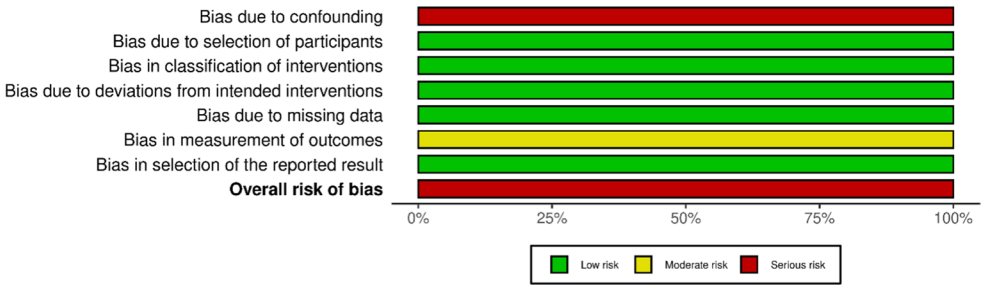


Fig. 3 Outcome-specific ROBINS-I assessment of the nonrandomized comparative study

allocation was related to baseline clinical characteristics and the resulting confounding was not statistically adjusted; we also noted additional concerns about outcome measurement. Detailed signaling-question responses and supporting judgments are provided in Supplementary Table S2.

Narrative synthesis of clinical outcomes

Surgical laser interventions in comparative studies

Three comparative studies evaluated lasers used during the surgical procedure for soft-tissue incision and/or osseous recontouring [11, 20, 21]. The interventions included diode and Er, Cr: YSGG laser systems and were compared with conventional scalpel-based or open-flap CL procedures.

Tianmitrapap et al. [11] compared flapless Er, Cr: YSGG laser-assisted CL with conventional open-flap surgery. Postoperative pain was lower in the laser group. In treated teeth, no significant between-group difference was observed in relative gingival-margin position, although the relative bone level was positioned more apically in the conventional-surgery group. In adjacent teeth, both the gingival margin and relative bone level were positioned more apically following conventional surgery.

Abdelhafez et al. [20] compared diode laser and conventional techniques using a split-mouth design. The groups showed no significant difference in postoperative pain or keratinized gingival width at 6 months. Clinical crown-length outcomes were also generally comparable, except for one tooth at the 6-month assessment, where the conventional group reported greater crown length.

Altayeb et al. [21] compared Er, Cr: YSGG laser-assisted flapless and open-flap crown lengthening. Mean gingival tissue rebound at 9 months was 0.26 ± 0.29 mm in the flapless group and 0.25 ± 0.32 mm in the open-flap group, with no significant between-group difference ($p = 0.742$). Greater rebound was reported in participants with a thick periodontal phenotype.

Regarding intraoperative conditions, Tianmitrapap et al. [11] reported reduced bleeding and improved surgical visibility with flapless Er, Cr: YSGG treatment compared with conventional flap surgery. Altayeb et al. [21] described minimal intraoperative bleeding during Er, Cr: YSGG-assisted procedures, while Abdelhafez et al. [20] reported improved operative-field visualization with diode-laser incision compared with scalpel incision. These findings were reported descriptively and were not assessed using a common standardized quantitative bleeding or visibility scale; therefore, they were not synthesized as quantitative comparative outcomes.

Overall, the comparative evidence suggested that surgical laser techniques may reduce postoperative pain in some settings and may produce periodontal and gingival-margin outcomes comparable with conventional procedures. However, the findings were not consistent across all studies, and differences in laser systems, surgical approaches, study designs, and follow-up periods limited direct comparison.

Adjunctive photobiomodulation

Two studies evaluated postoperative PBM as an adjunct to conventional osseous CL surgery [22, 23].

Sadighi et al. [22] evaluated an 860-nm diode laser against placebo treatment. No significant difference in postoperative pain was observed on Days 1 or 7, whereas pain intensity was significantly lower in the PBM group on Day 3. Analgesic consumption did not differ significantly between the groups.

Golpasandhagh et al. [23] evaluated dual-wavelength PBM using 980- and 810-nm diode lasers in a split-mouth design. The PBM-treated sites demonstrated significantly lower postoperative pain and lower analgesic consumption than the placebo-treated sites.

Taken together, adjunctive PBM showed potential benefits for early postoperative pain control. However, the timing and consistency of these effects varied between studies, and the available evidence was limited to two small trials using different irradiation protocols.

Overall direction and consistency of findings

The direction of the findings varied by clinical application of the laser. Surgical laser interventions generally demonstrated outcomes comparable with conventional surgery and showed possible advantages in postoperative pain and gingival-margin management in some studies. Adjunctive PBM was associated with reduced early postoperative pain in both included studies, although the benefit was not observed consistently at all assessment times.

Small sample sizes, variation in laser wavelengths and treatment parameters, differences between flapless and open-flap techniques, and inconsistent outcome assessment and follow-up periods limit interpretation of these findings. Accordingly, the available evidence does not permit identification of a single superior laser system or treatment protocol.

Discussion

This systematic review with narrative synthesis evaluated laser-based interventions in osseous crown-lengthening (CL) procedures by synthesizing evidence from five clinical studies and relevant supporting literature. Because the included studies differed in laser systems, intervention purposes, surgical approaches, comparators, and outcome measures, we considered surgical laser interventions and adjunctive PBM separately. The discussion is structured into four clinically relevant domains: (A) intraoperative outcomes, (B) postoperative pain and patient comfort, (C) healing and periodontal health, and (D) esthetic stability and long-term outcomes. Within each domain, findings from the five included studies are discussed first, while reviews, case reports, case series, and mechanistic literature outside the eligibility criteria are identified explicitly as contextual supporting evidence and are not used to infer comparative effectiveness. The findings are interpreted according to their clinical application, consistency, and methodological limitations.

Clinical interpretation of the evidence

Intraoperative outcomes: Visibility, hemostasis

From a surgical perspective, the intraoperative environment plays a critical role in procedural accuracy and clinician control. Several included studies descriptively reported favorable intraoperative conditions with laser-assisted procedures. Tianmitrapap et al. [11] reported reduced intraoperative bleeding and improved surgical visibility with flapless Er, Cr: YSGG-assisted crown lengthening, Altayeb et al. [21] described minimal intraoperative bleeding during Er, Cr: YSGG-assisted procedures, and Abdelhafez et al. [20] reported improved operative-field visualization with diode-laser incision compared with scalpel incision. However, these outcomes were not assessed using standardized quantitative bleeding or visibility scales across the studies. Accordingly, these observations should be interpreted as potential intraoperative advantages rather than established comparative effects.

Contextual evidence outside the included studies has highlighted potential technical limitations of flapless laser-assisted approaches. In a case series, McGuire and Scheyer [24] described complications related to “blind” osseous resection, including imprecise bone removal, irregular trough formation, and root-surface damage after subsequent flap reflection. Because this report was not an included comparative study, these observations are considered contextual rather than evidence of comparative effectiveness.

Operative time was not reported consistently enough across the included studies to support a review-level conclusion. Contextual literature outside the included evidence has reported both shorter and longer procedure times with laser-assisted crown lengthening, depending on the laser system and surgical technique [6, 12, 15, 25, 26]. These external observations were considered only as contextual information and were not used to infer comparative effectiveness within the present review.

An important interpretive limitation is that the effects of laser use cannot always be separated from those of flap design. In Tianmitrapap et al. [11], the flapless Er, Cr: YSGG approach was compared with conventional open-flap surgery; therefore, differences in bleeding, visibility, postoperative discomfort, or gingival-margin behavior may reflect the combined effects of laser use and the less invasive flapless surgical approach rather than the laser alone. Similarly,

Altayeb et al. [21] compared flapless and open-flap protocols within an Er, Cr: YSGG-assisted setting, meaning that observed differences primarily inform the influence of surgical approach rather than an isolated laser effect. These design features limit causal attribution of apparent clinical advantages specifically to laser technology.

Postoperative pain and patient comfort

While intraoperative parameters primarily reflect the clinician's experience, postoperative pain and patient comfort represent important outcomes from the patient's perspective. Pain has therefore been a frequently assessed endpoint in studies of both laser-assisted surgery and adjunctive PBM following osseous CL.

Among the included studies that assessed postoperative pain, laser-assisted surgery or adjunctive PBM reduced pain at selected assessment times; however, the findings were not consistent across studies. Tianmitrapap et al. [11] reported significantly lower pain scores on Day 1 in patients treated with flapless Er, Cr: YSGG lasers compared with conventional open-flap surgery. In contrast, Abdelhafez et al. [20] found no statistically significant difference in postoperative pain between the diode-laser and scalpel groups, indicating no clear analgesic advantage for the diode laser in that trial. Supporting literature has also emphasized the importance of adequate cooling during erbium-laser application because excessive thermal exposure may contribute to tissue damage and postoperative discomfort [27]. Furthermore, a systematic review by Hamadah et al. [14] reported increased immediate postoperative pain with Nd: YAG lasers and discussed deeper thermal penetration and collateral tissue effects as possible contributing mechanisms.

Contextual evidence from studies outside the systematic review has also described favorable patient-reported outcomes with erbium lasers. A systematic review by Elafifi Ebeid et al. [15] reported minimal postoperative inflammation and low pain scores following flapless Er: YAG and Er, Cr: YSGG CL, consistent with reduced tissue trauma. Similarly, a case series by Fekrazad et al. [28] reported low postoperative pain and few complications following Er, Cr: YSGG-assisted CL.

Adjunctive PBM has also demonstrated potential benefits. Golpasandhagh et al. [23] reported that dual-wavelength diode PBM (980/810 nm) significantly reduced pain scores and analgesic consumption on postoperative Days 1, 2, and 4 following conventional osseous CL. Similarly, Sadighi et al. [22] observed reduced pain at Day 3 using an 860 nm diode laser, although no differences were noted at Days 1 or 7 or in analgesic intake. These findings should be interpreted separately from studies of surgical lasers because PBM was used after conventional surgery rather than as a replacement for the conventional surgical procedure.

Healing and periodontal health

Although postoperative pain reflects short-term outcomes, healing quality and periodontal health are critical determinants of long-term success. The included studies primarily provide clinical evidence regarding tissue response, while mechanistic and histological considerations are derived mainly from supporting literature.

Tianmitrapap et al. [11] and Altayeb et al. [21] reported uneventful healing and absence of adverse tissue reactions following flapless laser-assisted CL, with outcomes comparable to conventional methods. Supporting literature has additionally described a limited zone of thermal alteration when Er: YAG lasers are used with adequate cooling [27].

Adjunctive PBM may also support postoperative healing. Mechanistic literature has proposed that PBM may influence fibroblast activity, angiogenesis, inflammatory signaling, and growth-factor pathways. These mechanisms were not directly evaluated in the included PBM trials and are therefore presented only as contextual biological rationale rather than as findings of this review [22, 23].

Supporting literature outside the included clinical studies has also reported favorable healing outcomes. A case report using Er: YAG lasers demonstrated stable periodontal conditions at 3- and 12-month follow-ups [25]. Aoki et al. [12] reported complication-free recovery comparable to conventional instrumentation but with reduced postoperative pain and edema.

Esthetic stability and long-term outcomes

The long-term success of CL depends on gingival-margin stability and overall esthetic outcomes. The included studies provide limited but generally favorable evidence regarding laser-assisted techniques. Tianmitrapap et al. [11] reported no significant between-group difference in relative gingival-margin position in treated teeth at 3 months. Relative bone level was positioned more apically in the conventional-surgery group, while in adjacent teeth both relative gingival-margin position and relative bone level were more apically positioned following conventional surgery. These findings should be interpreted cautiously because the study was non-randomized and was judged to have a serious risk of bias, primarily due to confounding related to treatment allocation. Altayeb et al. [21] reported stable gingival margins from 3 to 9 months in both open-flap and flapless laser groups, with only minor, phenotype-dependent rebound. Abdelhafez et al. [20] reported stable outcomes up to 6 months, along with increased patient satisfaction over time.

Contextual evidence from previous systematic reviews has reported generally comparable gingival-margin stability between laser and conventional techniques and has

emphasized the importance of an adequate maturation period before definitive restoration [15, 29]. Elafifi Ebeid et al. [15] further suggested that flapless erbium-laser techniques may improve procedural predictability by avoiding flap repositioning and suturing.

In contrast, conventional CL is associated with biologically driven gingival rebound. Majzoub et al. [3] and Altayeb et al. [30] noted that tissues tend to re-establish supracrestal dimensions, leading to coronal migration of the gingival margin. Hempton and Dominici [31] reported rebound of approximately 3 mm following surgery. Arora et al. [7] demonstrated that this effect is technique-sensitive and influenced by flap positioning, while Domínguez et al. [32] emphasized the importance of the distance between the gingival margin and alveolar bone. Pilalas et al. [29] reported that most remodeling occurs within 3 months but may continue up to 6 months. Accordingly, a maturation period of 3–6 months is generally recommended to ensure stability [30, 31].

Limitations

This systematic review has several limitations. First, only five clinical studies were included, most with relatively small sample sizes and heterogeneous designs comprising randomized and nonrandomized comparative studies. Second, substantial clinical and methodological heterogeneity was present in laser type, wavelength, energy settings, surgical technique, comparator, treatment indication, outcome definition, and follow-up duration. Surgical laser interventions and postoperative photobiomodulation also represented distinct clinical applications, precluding meaningful quantitative pooling. Third, two included studies used split-mouth designs, in which observations from treatment sides within the same participant are correlated. One study explicitly used paired and repeated-measures analyses, whereas the extent to which within-participant correlation was accounted for in all reported analyses of the other split-mouth trial was not always clear from the published report. This may affect the precision of some reported estimates and should be considered when interpreting these findings. Fourth, outcome reporting was inconsistent, particularly for postoperative pain, healing, gingival-margin stability, operative time, patient satisfaction, and adverse events, which limited direct comparison across studies. Fifth, the outcome-specific RoB 2 assessments identified a high risk of bias for one assessed result and some concerns for the assessed results from the remaining three randomized studies; none of the assessed randomized-trial results were judged to have a low risk of bias. The single nonrandomized comparative study was judged to have a serious risk of bias using ROBINS-I, primarily because of confounding related to treatment allocation. These methodological limitations reduce confidence in the comparative interpretation of the available findings. The restriction to English-language publications may have introduced language bias. ScienceDirect was included as an additional electronic literature source alongside the bibliographic databases; however, its search functionality and indexing differ from those of conventional bibliographic databases, which may have influenced retrieval. In addition, several included studies did not report formal effect estimates or confidence intervals for all outcomes, limiting assessment of effect magnitude and precision despite the additional numerical data provided in Supplementary Table S4. Furthermore, no formal certainty-of-evidence framework such as GRADE was applied. Finally, the narrative synthesis did not provide a pooled quantitative estimate of treatment effect; therefore, the findings should be interpreted as a summary of the direction and consistency of the available evidence rather than as evidence of definitive clinical superiority.

Conclusion

Within the limitations of the available evidence, some studies descriptively reported reduced intraoperative bleeding and improved surgical visibility with surgical laser techniques; however, these outcomes were not assessed using standardized quantitative measures. Periodontal and gingival-margin outcomes were generally comparable with conventional surgical approaches, and the current evidence does not demonstrate a consistent clinical advantage of surgical laser techniques. Adjunctive PBM may reduce early postoperative pain at selected assessment times; reduced analgesic consumption was reported in only one of the two PBM studies. Laser-based interventions may serve as surgical alternatives or postoperative adjuncts in selected clinical settings; however, the current evidence remains too limited and heterogeneous to support definitive clinical recommendations or identify an optimal laser system or treatment protocol. Future studies should evaluate surgical laser interventions and adjunctive PBM as distinct clinical applications and use rigorous prospective designs with transparent random-sequence generation and allocation concealment. Primary outcomes and assessment time points should be predefined, outcome assessors should be blinded when feasible, and complete laser dosimetry and procedural parameters should be reported. Split-mouth trials should use statistical methods that account for within-participant correlation, and treatment effects should be reported with appropriate measures of variability and confidence intervals. Prospective protocol registration before participant recruitment and sufficiently long follow-up are also needed to improve methodological transparency and assess periodontal and gingival-margin stability.

Glossary

CL	Crown lengthening
EGD	Excessive gingival display
PBM	Photobiomodulation; the therapeutic application of low-intensity light to modulate pain, inflammation, and tissue healing without surgical tissue ablation
LLLT	Low-level laser therapy; an earlier term commonly used for photobiomodulation
Er	YAG: Erbium-doped yttrium aluminum garnet laser
Er,Cr	YSGG: Erbium, chromium-doped yttrium scandium gallium garnet laser
Nd	YAG: Neodymium-doped yttrium aluminum garnet laser
CO_2	Carbon dioxide laser
GaAlAs	Gallium aluminum arsenide laser
PICOS	Population, intervention, comparison, outcomes, and study design
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RoB 2	Revised Cochrane Risk of Bias tool
SWiM	Synthesis Without Meta-analysis
ROBINS-I	Risk Of Bias In Non-randomized Studies of Interventions
VAS	Visual analog scale

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate This systematic review analyzed data from previously published studies and did not involve the direct recruitment of human participants or collection of identifiable personal information. Therefore, ethical approval and informed consent were not required.

Competing interests The authors declare no competing interests.

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